Day 28 Classwork

Tuesday, April 8, 2025 9:20 PM

Day 28 Classwork

Algebra 2 Honors Name: _____

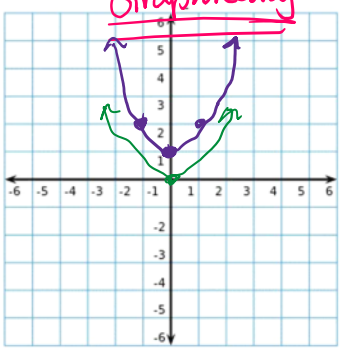
Day 28 | 8.2.2 & 8.2.3 | Complex Numbers

Warm-Up

What are the roots of this polynomial: $y = x^2 + 1$

x-ints

Graphically



Algebraically

$0 = x^2 + 1$

$-1 \quad -1$

$\sqrt{-1} = \sqrt{x^2}$

$\pm\sqrt{-1} = x$

no solution $= x$

$a = 1$

$b = 0$

$c = 1$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \pm\sqrt{-1}$

Video

1. What does the Fundamental Theorem of Algebra state?

highest degree (n) = # of roots

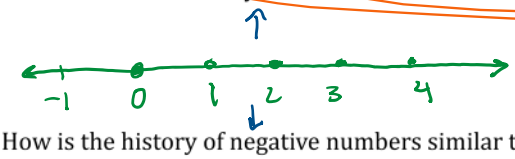
# of real + # imaginary roots = degree

2. What is another way of calling imaginary numbers?

Lateral #'s

Real #

3. How has our number system evolved over time?



4. How is the history of negative numbers similar to the history of imaginary numbers?

Neg #'s not viewed as needed/important

Didn't belief

8-97 Use the method described in video to plot points to represent the six numbers below.

A. $2 + 5i$

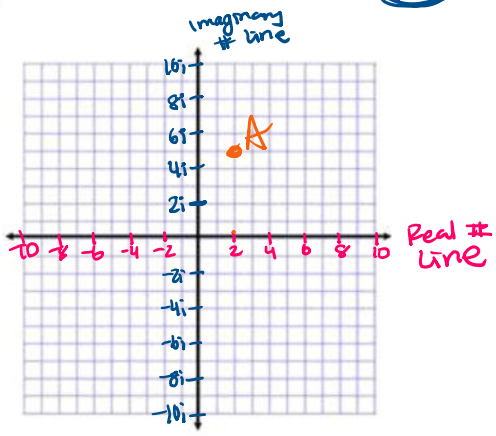
B. $6 - i$

C. $-5 - 3i$

D. 4

E. $7i$

F. $-4 + 2i$



$a \pm bi$

real # imaginary #

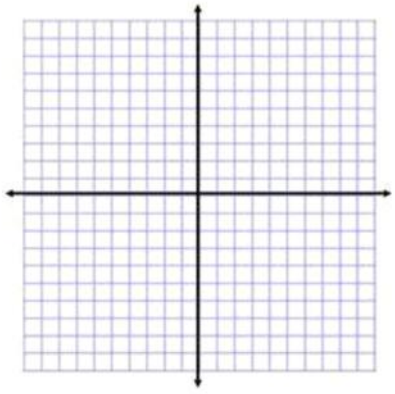
8-98. On a new set of **complex axes**, locate points representing all of the following complex numbers.

a. Using **STARS:** $3 + 4i$, $3 - 4i$, $-3 + 4i$, and $-3 - 4i$

b. Using **CIRCLES:** 5, -5, $5i$, and $-5i$

c. How far from $(0, 0)$ is $-3 - 4i$?

d. How far from $(0, 0)$ is $5i$?



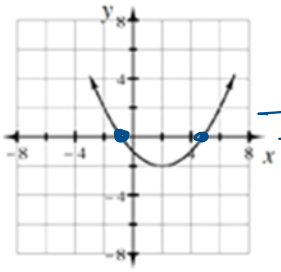
8-99. On the real number line, the distance from 0 to a point on the line is defined as the absolute value of the number. Similarly, in the **complex plane** (the plane defined by a set of complex axes), the **absolute value** of a complex number is its distance from zero or the origin $(0, 0)$. In the previous problem, the absolute value of all of those complex numbers was 5.

a. What is the absolute value of $-8 + 6i$?

b. What is the absolute value of $a + bi$?

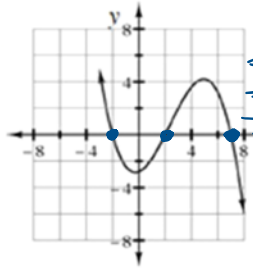
8-100. Based on the following graphs, how many *real* roots does each polynomial function have?

a.



real roots = 2
imaginary roots = 0
total degree = 2

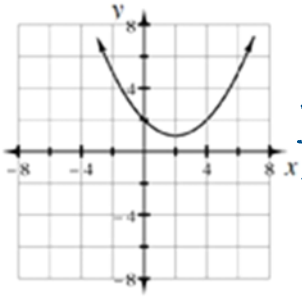
b.



real roots = 3
imaginary roots = 0
total degree = 3

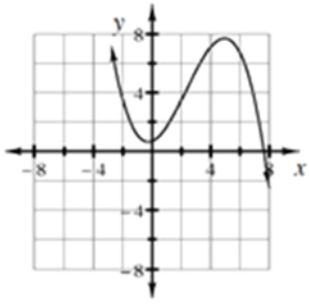
Graphs (a) and (b) above have been vertically shifted to create graphs (c) and (d) shown below. How many *real* roots does each of these new polynomial functions have?

c.



real 0
imaginary 2
total 2

d.



real 1
imaginary 2
total 3

8-65. In the 1500s, an Italian mathematician named Rafael Bombelli invented the imaginary number, which is now called i .

$$\sqrt{-1} = i \text{ implies that } i^2 = -1.$$

For example, the value of $\sqrt{-16} = \sqrt{16 \cdot -1} = \sqrt{16} \cdot \sqrt{-1} = 4i$

Use the definition of i to rewrite each of the following expressions:

a. $\sqrt{-4}$ $= \sqrt{-1} \cdot \sqrt{4} = \boxed{2i}$

b. $(2i)(3i)$ $= 2 \cdot 3 \cdot i \cdot i = 6i^2 = 6 \cdot -1 = \boxed{-6}$

6

c. $(2i)(-5i)$ $= -10i^2 = -10 \cdot -1 = \boxed{10}$

d. $\sqrt{-25}$ $= \sqrt{25} \cdot \sqrt{-1} = \boxed{5i}$

$i^2 = -1$

$\sqrt{-1} \cdot \sqrt{-1}$

-1

8-66. Here is the function $y = x^2 - 4x + 5$.

a. Based on the Fundamental Theorem of Algebra, how many roots should the equation have?

b. Use the Quadratic Formula to find the roots of the given equation.

8-79. Find the roots of each of the following quadratic functions by solving for x when y = 0.

$y = (x + 5)^2 + 9$

$0 = (x+5)^2 + 9$

$-9 \qquad -9$

$\sqrt{-9} = \sqrt{(x+5)^2}$

$\pm\sqrt{-9} = x+5$

$\pm 3i = x+5$

$\boxed{-5 \pm 3i = x}$

$y = x^2 - 4x + 9$

$a=1 \quad b=-4 \quad c=9$

$x = \dfrac{-b \pm \sqrt{b^2-4ac}}{2a} = \dfrac{4 \pm \sqrt{16-4(1)(9)}}{2}$

$= \dfrac{4 \pm \sqrt{16-36}}{2}$

$= \dfrac{4 \pm \sqrt{-20}}{2}$

$= \boxed{2 \pm \sqrt{5}i}$

$= \dfrac{4 \pm 2\sqrt{5}i}{2}$

$\sqrt{-20}$

$\sqrt{20}\ \sqrt{-1}$

$\sqrt{5}\ \sqrt{4}$

$\sqrt{5}\ 2$

8-86. For each pair of numbers below, find a quadratic equation that has these numbers as solutions.

a. $\frac{3}{4}$ and -5

b. $x = 3i$ and $x = -3i$

$\boxed{(x-r_1)(x-r_2)}$

$(x-3i)(x+3i)$

	x	$3i$
x	x^2	$3ix$
$-3i$	$-3ix$	$-9i^2 = -9 \cdot -1 = 9$

$\boxed{x^2+9}$

c. $5 + 2i$ and $5 - 2i$

$(x-(5+2i))(x-(5-2i))$

$(x-5-2i)(x-5+2i)$

	x	-5	$+2i$
x			
-5			
$-2i$			

d. $-3 + \sqrt{2}$ and $-3 - \sqrt{2}$